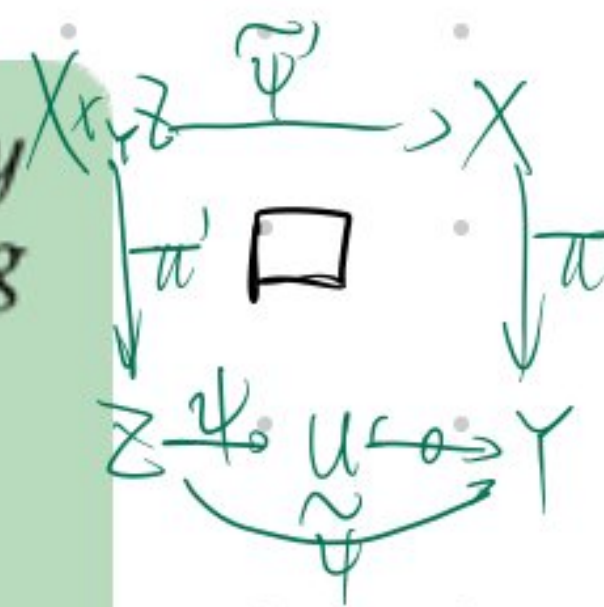


Vakil - Chapter 25. Reading Note.

25.1.6. Cohomology and Base Change Theorem. — Suppose π is proper, Y is locally Noetherian, $\mathcal{F}$ is coherent over X and flat over Y , and ϕ_q^p is surjective. Then the following hold.

$$R^i \pi_* \mathcal{F} \otimes_{k(q)} \rightarrow H^i(X_q, \mathcal{F}|_{X_q}).$$



(i) There is an open neighborhood U of q such that for any $\psi: Z \rightarrow U$, ϕ_Z^p is an isomorphism. In particular, ϕ_q^p is an isomorphism.

(ii) Furthermore,

(a) ϕ_q^{p-1} is surjective (hence an isomorphism by (i)) if and only if

(b) $R^p \pi_* \mathcal{F}$ is locally free in some open neighborhood of q (or equivalently, $(R^p \pi_* \mathcal{F})_q$ is a free $\mathcal{O}_{Y,q}$ -module, Exercise 14.3.E).

(These then imply that $h^p(X_r, \mathcal{F}|_{X_r})$ is constant for r in an open neighborhood of q .)

Fact: can be replaced by "finitely presented".

$$\psi^* R^p \pi_* \mathcal{F} \xrightarrow{\phi_Z^p} R^p \pi'_* (\psi^* \mathcal{F})$$

Applications:

25.1.B. EXERCISE. Use Theorem 25.1.6 to give a second solution to Exercise 24.3.E. (This is a big weapon to bring to bear on this problem, but it is still enlightening; your original solution to Exercise 24.3.E may foreshadow the proof of the Cohomology and Base Change Theorem 25.1.6.)

24.3.E. EXERCISE. Suppose $\pi: X \rightarrow Y$ is a finite flat morphism of locally Noetherian schemes, and $\mathcal{F}$ is a finite rank locally free sheaf on X .

(a) Show that $\pi_* \mathcal{F}$ is a finite rank locally free sheaf on Y .

(b) If Y is irreducible with generic point η , the degree of π above η is n , and $\mathcal{F}$ is locally free of rank r , show that $\pi_* \mathcal{F}$ is locally free of rank nr . (This may be trickier than it looks.)

Verify the condition in Cohomology and Base Change Theorem:

Surjectivity of ϕ_q^p for any $q \in Y$:

$X \rightarrow Y$ finite, hence affine. Choose $q \in \text{Spec } A$ & $\pi^{-1}(\text{Spec } A) = \text{Spec } B$ & $\mathcal{F}|_{\text{Spec } B} \cong \tilde{M}$.
 $p=0$, $\pi_* \mathcal{F} \otimes_{k(q)} \rightarrow H^0(X_q, \mathcal{F}|_{X_q})$ is actually $M \otimes_A k(q) \xrightarrow{\sim} M \otimes_B (B \otimes_A k(q))$
 $p>0$, well-known that $R^i \pi_* \mathcal{F} = 0$ ($\because \pi$ affine).
 & $H^i(X_q, \mathcal{F}|_{X_q}) = 0$ ($\because X_q \cong \text{Spec } (B \otimes_A k(q))$ affine).

$\therefore \phi_q^p$ are actually isomorphisms at this situation!

Apply Coh. and BC Theorem

(i) apply for $p=0$ and all q

By ϕ_q^0 surjective, $\pi_* \mathcal{F} = R^0 \pi_* \mathcal{F}$ is locally free.

$$\text{rank } \pi_* \mathcal{F} = \dim_{k(q)} \pi_* \mathcal{F} \otimes_{k(q)} = \dim_{k(q)} H^0(X_q, \mathcal{F}|_{X_q}).$$

by ϕ_q^0 isomorphism

$X_q \cong \text{Spec } C$ with C : finite dimensional $k(q)$ -alg., hence $C = C_1 \times C_2 \times \dots \times C_s$

$$H^0(X_q, \mathcal{F}|_{X_q}) = P_1 \oplus \dots \oplus P_s \text{ with } \mathcal{F}|_{\text{Spec } C_i} \cong \tilde{P}_i, P_i: \text{free } C_i\text{-module}$$

($\because \mathcal{F}|_{X_q}$ locally free $\Rightarrow$ fin. proj. $\Rightarrow$ free.)

Proposition 10C.1. Let R be a ring. The following are equivalent:
 (1) R is Artinian.
 (2) R is Noetherian and $\dim(R) = 0$.
 (3) R has finite length as a module over itself.
 (4) R is a finite product of Artinian local rings.
 (5) R is Noetherian and $\text{Spec}(R)$ is a finite discrete topological space.
 (6) R is a finite product of Noetherian local rings of dimension 0.
 (7) R is a finite product of Noetherian local rings R_i with $d(R_i) = 0$.
 (8) R is a finite product of Noetherian local rings R_i whose maximal ideals are nilpotent.
 (9) R is Noetherian, has finitely many maximal ideals and its Jacobson radical ideal is nilpotent.
 (10) R is Noetherian and there are no strict inclusions among its primes.

Artin

Artinian local ring

$$\dim_{k(q)} H^i(X_q, \mathcal{F}|_{X_q}) = \sum_{i=1}^r \dim_{k(q)} C_i \cdot \dim_{C_i} P_i = \left(\sum_{i=1}^r \dim_{k(q)} C_i \right) \cdot r = r \cdot \deg \pi. \quad \square$$

degree of π above q .

References:

Lemma 00NX. Let R be a ring and let M be an R -module. The following are equivalent

- (1) M is finitely presented and R -flat,
- (2) M is finite projective,
- (3) M is a direct summand of a finite free R -module,
- (4) M is finitely presented and for all $\mathfrak{p} \in \text{Spec}(R)$ the localization $M_{\mathfrak{p}}$ is free,
- (5) M is finitely presented and for all maximal ideals $\mathfrak{m} \subset R$ the localization $M_{\mathfrak{m}}$ is free,
- (6) M is finite and locally free,
- (7) M is finite locally free, and
- (8) M is finite, for every prime $\mathfrak{p}$ the module $M_{\mathfrak{p}}$ is free, and the function

$$\rho_M : \text{Spec}(R) \rightarrow \mathbb{Z}, \quad \mathfrak{p} \mapsto \dim_{k(\mathfrak{p})} M \otimes_R k(\mathfrak{p})$$

is locally constant in the Zariski topology.

Lemma 05JM. Let $X = \text{Spec}(R)$ be an affine scheme. Let $\mathcal{F} = \widetilde{M}$ for some R -module M . The quasi-coherent sheaf $\mathcal{F}$ is a (finite) locally free $\mathcal{O}_X$ -module if and only if M is a (finite) locally free R -module.

24.3.5. Important Theorem (flat = free = projective for finitely presented modules over local rings). — Suppose $(A, \mathfrak{m})$ is a local ring (not necessarily Noetherian), and M is a finitely presented A -module. Then the following are equivalent.

- (a) M is free.
- (b) M is projective.
- (c) M is flat.
- (d) $\text{Tor}_1^A(M, A/\mathfrak{m}) = 0$.

25.1.C. EXERCISE. Suppose $\pi: X \rightarrow Y$ is proper, Y is locally Noetherian, and $\mathcal{F}$ is a coherent sheaf on X , flat over Y . Suppose further that $H^p(X_q, \mathcal{F}|_{X_q}) = 0$ for some $q \in Y$. Show that there is an open neighborhood U of q such that $(R^p \pi_* \mathcal{F})|_U = 0$.

pf. Apply Cohomology and Base Change Theorem for q, p .

$$\xRightarrow{(i)} \phi_q^p : R^p \pi_* \mathcal{F}|_q \xrightarrow{\cong} H^p(X_q, \mathcal{F}_q) = 0.$$

coherent

$\therefore$ By Nakayama, $(R^p \pi_* \mathcal{F})_q = 0$ stalk zero.

$\therefore \exists U \ni q$ s.t. $(R^p \pi_* \mathcal{F})|_U = 0$. $\square$

(Recall the details: $M: A$ -module, $M_{\mathfrak{p}} = 0$, f_i gen. by $m_1, \dots, m_s$. $f_i m_i = 0$ for $f_i \in \mathfrak{p}$.)
Consider $D(f_1, \dots, f_s) \in \text{Spec } A$

difference

25.1.D. EXERCISE. Suppose $\pi: X \rightarrow Y$ is proper, Y is locally Noetherian, and $\mathcal{F}$ is a coherent sheaf on X , flat over Y . Suppose further that $H^p(X_q, \mathcal{F}|_{X_q}) = 0$ for all $q \in Y$. Show that the $(p-1)$ st cohomology commutes with arbitrary base change: ϕ_Z^{p-1} is an isomorphism for all $\psi: Z \rightarrow Y$.

pf. By 25.1.C. $R^p \pi_* \mathcal{F} = 0$. automatically locally free.

Apply Cohomological and Base Change (ii):

for p , any q . ϕ_q^{p-1} are surj.

Then apply (i). for any q , $\exists U_q \ni q$ s.t. $\forall Z_q \rightarrow U_q$, $\phi_{Z_q}^{p-1}$ isom.

for any $Z \rightarrow Y$. $\{U_q\}$ covers Y . $\{U_q \times_Y Z\}$ covers Z .

$\phi_{Z_q}^{p-1}$ isom "gluing together" ϕ_Z^{p-1} isom. $\square$

(is just $\psi^* R^p \pi_* \mathcal{F} \xrightarrow{\cong} R^p \pi_* (\psi^* \mathcal{F})$)
 $\forall \psi: Z \rightarrow Y$.

25.1.E. EXERCISE. Suppose π is proper, Y is locally Noetherian, and $\mathcal{F}$ is a coherent sheaf on X flat over Y . Suppose further that $R^p\pi_*\mathcal{F} = 0$ for $p \geq p_0$. Show that $H^p(X_q, \mathcal{F}|_{X_q}) = 0$ for all $q \in Y$, $p \geq p_0$.

pf. For any $q \in Y$, $H^p(X_q, \mathcal{F}|_{X_q}) = 0 \quad \forall p \geq p_1$.
 depends on q !
 large enough
 by Grothendieck's Vanishing.
 By (ii) $R^p\pi_*\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y,q} \xrightarrow{\sim} H^p(X_q, \mathcal{F}|_{X_q})$ isom.
 $(\because R^{p+1}\pi_*\mathcal{F}$ automatically locally free).
 By (i) again $\dots \dots \dots p-2 \dots \dots \dots p-1 \dots \dots \dots$
 $\implies H^p(X_q, \mathcal{F}|_{X_q}) = 0 \quad \forall p \geq p_0. \quad q \in Y. \quad \square$

25.1.F. EXERCISE. Suppose π is proper, Y is locally Noetherian, and $\mathcal{F}$ is a coherent sheaf on X , flat over Y . Suppose further that Y is reduced. Show that there exists a dense open subset U of Y such that ϕ_Z^p is an isomorphism for all $\psi: Z \rightarrow U$ and all p . (Hint: find suitable open neighborhoods of the generic points of Y . See Exercise 24.1.M and the paragraph following it.)

24.1.9. Theorem (cohomology commutes with flat base change). — Suppose

$$\begin{array}{ccc} X' & \xrightarrow{\rho'} & X \\ \pi' \downarrow & & \downarrow \pi \\ Y' & \xrightarrow{\rho} & Y \end{array}$$

is a Cartesian diagram, and π (and thus π') is quasicompact and separated (so higher pushforwards of quasicoherent sheaves by π and π' exist, as described in §18.7). Suppose also that ρ is flat, and $\mathcal{F}$ is a quasicoherent sheaf on X . Then the natural “push-pull” morphisms (Exercise 18.7.B(a)) $\rho^*(R^i\pi_*\mathcal{F}) \rightarrow R^i\pi'_*(\rho'^*\mathcal{F})$ are isomorphisms.

24.1.M. EXERCISE. Prove Theorem 24.1.9. Hint: Exercise 18.7.B(b) is the special case where ρ is affine. Extend it to the quasicompact and separated case using the same idea as the solution to Exercise 14.4.C (which used Exercise 6.2.G). Your proof of the case $i = 0$ will only need a quasiseparated hypothesis in place of the separated hypothesis.

18.7.B (b) (cohomology commutes with affine flat base change) If $\psi: Y' \rightarrow Y$ is an affine morphism, and for a cover $\text{Spec } A_i$ of Y , where $\psi^{-1}(\text{Spec } A_i) = \text{Spec } B_i$, B_i is a flat A_i -algebra (§1.5.13: $\otimes_{A_i} B_i$ is exact), and the diagram in (a) is a Cartesian diagram, show that the push-pull map of (a) is an isomorphism. (Exercise 18.2.H was a special case of this exercise. You can likely generalize this to non-affine morphisms — and thus show that cohomology commutes with flat base change (Theorem 24.1.9) — but we wait until Chapter 24 to discuss flatness at length.)

$$\begin{array}{ccc} X' & \xrightarrow{\psi'} & X \\ \pi' \downarrow & & \downarrow \pi \\ Y' & \xrightarrow{\psi} & Y \end{array}$$

$$\psi^*(R^i\pi_*\mathcal{F}) \longrightarrow R^i\pi'_*(\psi')^*\mathcal{F}$$

14.4.C. ESSENTIAL EXERCISE (COROLLARY TO EXERCISE 6.2.G: " $\pi_* : \text{QCoh}_X \rightarrow \text{QCoh}_Y$ FOR QCQS π "). Suppose $\pi: X \rightarrow Y$ is a quasicompact quasiseparated morphism, and $\mathcal{F}$ is a quasicoherent sheaf on X . Show that $\pi_* \mathcal{F}$ is a quasicoherent sheaf on Y . (For counterexamples without the quasicompact or quasiseparated hypotheses, see [Stacks, tag 078C].)



A useful special case is where Y' is the generic point of a reduced component of Y . In other words, in light of Exercise 24.1.G(a), the stalk of the higher pushforward of $\mathcal{F}$ at the generic point is the cohomology of $\mathcal{F}$ on the fiber over the generic point.

(The paragraph after 24.1.U).



By cohomology commutes by fiber base change (24.1.9).

isom. $(R^p \pi_* \mathcal{F}) \otimes k(\eta) \xrightarrow{\sim} R^p \pi'_*(\mathcal{F}|_{X_\eta}), \forall p \geq 0.$
(in particular, ϕ_η^p is surj.).

So by Cohomology and Base Change $\Rightarrow \exists U_p \ni \eta$ s.t. such that.

$\forall Z \rightarrow U_p, \phi_Z^p$ is isomorphism.

consider $U_p = Y \setminus \bigcup_{\eta \text{ generic points of } Y} (\overline{\{\eta\}} \setminus U_p^\eta)$ dense open in Y . $U_p \cap \overline{\{\eta\}} \in U_p^\eta$
closed in $\overline{\{\eta\}}$, hence in Y .

Y locally finite.

closed in Y by topological reason

相当于 reduced to Y is irreducible.

A family of subsets $\{A_i\}_{i \in I}$ of a topological space Y is called locally finite if for every $y \in Y$, there exists an open neighborhood $V \ni y$ such that $V \cap A_i \neq \emptyset$ for only finitely many $i \in I$.

This condition implies that an arbitrary union of closed sets is closed. Indeed, suppose each Z_i is closed and $\{Z_i\}$ is locally finite. Take $y \notin Z := \bigcup Z_i$. Choose an open neighborhood $V \ni y$ which meets only, say, $Z_{i_1}, \dots, Z_{i_n}$. Then $V' := V \setminus (Z_{i_1} \cup \dots \cup Z_{i_n})$ is open and contains y . Moreover V' meets no Z_i ; the other Z_i 's did not meet V in the first place. Thus $V' \subseteq Y \setminus Z$. So $Y \setminus Z$ is open, hence $\bigcup Z_i$ is closed.

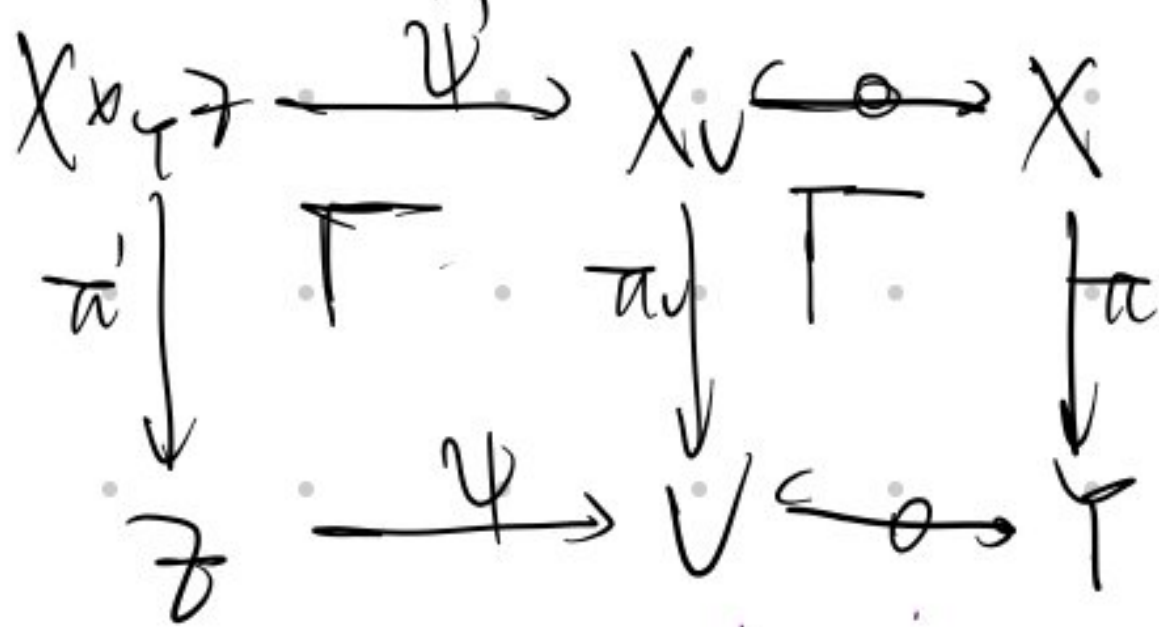
(We can use Grothendieck's Vakil's because $\dim Y$ may be $+\infty$)

(Warning: now we only choose dense open U_p for any p . but the exercise needs us find a unified U . while $\bigcap U_p$ may not be open)

(method: directly choose another dense open $V \subseteq Y$ s.t. for p large enough ($p \geq p_0$)) $\forall Z \rightarrow V, \phi_Z^p$ is actually $0 \rightarrow 0$. then consider $U = (\bigcap_{p \geq p_0} U_p) \cap V$

By above discussion on reduced to Y irreducible". WMA Y irreducible

Claim: $\exists \text{Gen } A = V$ (dense) open in Y , such that $\forall Z \xrightarrow{\psi} V$ and $p \geq p_0$. & $p_0 \in \mathbb{N}^*$



$$\begin{cases} R^p \pi_{V*} \mathcal{F}|_{X_V} = 0. \\ R^p \pi'_*(\psi'^* \mathcal{F}|_{X_V}) = 0. \end{cases} \quad (\text{immediately from the following}). \quad \text{--- ①}$$

reason: By affinities of V and π proper, X_V has finite affine covering.

Write $X_V = \bigcup_{i \in I} W_i$. $W_i \cong \text{Spec } C_i$.

check $\mathbb{D}$ affine locally: want to show, $\exists p_0 \in \mathbb{N}^*$, $\forall T = \text{Spec } B \in \mathbb{Z}$.

$$\Gamma(T, R^p \pi_* (\psi^* \mathcal{F}|_{X_V})) = H^p(X \times_V T, \psi^* \mathcal{F}|_{X_V}) = 0.$$

In fact, choose $p_0 = \#I$.

$\{W_i \times_V T\}$ affine covering of $X \times_V T$.
 $\text{Spec } C_i \otimes_A B$.

By {
18.2.2. Theorem/Definition. — Our standing assumption is that X is quasicompact and separated. $H^i_{\mathcal{U}}(X, \mathcal{F})$ is independent of the choice of (finite) cover $\{U_i\}$. More precisely, for any two covers $\{U_i\} \subset \{V_i\}$, the maps $H^i_{\{V_i\}}(X, \mathcal{F}) \rightarrow H^i_{\{U_i\}}(X, \mathcal{F})$ induced by the natural map of Čech complexes (18.2.1.1) are isomorphisms. Define the Čech cohomology group $H^i(X, \mathcal{F})$ to be this group.
23.5.1. Theorem. — Suppose X is a quasicompact separated scheme, and $\mathcal{F}$ is a quasicoherent sheaf. Then the Čech cohomology of $\mathcal{F}$ agrees with the derived functor cohomology of $\mathcal{F}$.

$\forall p \geq p_0, H^p = 0$
 Since the length of the complex $\leq p_0$.

$$U = \bigcap_{p \leq p_0} U_p \cap V. \quad \square$$

25.1.G. EXERCISE. Suppose $\pi: C \rightarrow S$ is a proper smooth morphism whose fibers are genus $g > 1$ curves (a “flat family of smooth genus g curves”). Suppose further that S is locally Noetherian.

(a) If $n > 1$, show that $\pi_* \Omega_{C/S}^{\otimes n}$ is a vector bundle of rank $(2n-1)(g-1)$, and that this construction commutes with any base change $S' \rightarrow S$ (with no Noetherian hypotheses on S'). More precisely, if

$$\begin{array}{ccc} C' & \xrightarrow{\rho'} & C \\ \pi' \downarrow & & \downarrow \pi \\ S' & \xrightarrow{\rho} & S \end{array}$$

Link: in fact can be removed, but using

different way to prove. hence 25.1.G generalized to $n \geq 1$.

is a Cartesian diagram, show that $\rho^* \pi_* \Omega_{C/S}^{\otimes n} \cong \pi'_* \Omega_{C'/S'}^{\otimes n}$.

(b) If S is reduced, show that $\pi_* \Omega_{C/S}$ is a vector bundle of rank g , and that this construction commutes with any base change $S' \rightarrow S$ (with no hypotheses on S).

pf. (a) do “More precisely” first.

Apply 25.1.D. only need to show for all $q \in Y$. $H^1(C_q, \Omega_{C_q/q}^{\otimes n}) = 0$.

By the condition, C_q is a smooth genus g curve over q .

By Serre duality. $h^1(C_q, \Omega_{C_q/q}^{\otimes n}) = h^0(C_q, \Omega_{C_q/q}^{\otimes (1-n)})$

Since $\deg \Omega_{C_q/q}^{\otimes (1-n)} = (1-n)(2g-2) \leq 0$. $\therefore h^0(C_q, \Omega_{C_q/q}^{\otimes (1-n)}) = 0$.

Hence $H^1(C_q, \Omega_{C_q/q}^{\otimes n}) = 0$.

In particular, $S' = \{q\}$. we have ϕ_q^0 is isomorphism.

By Cohomology and Base Change (ii) from ϕ_q^1 automatically surj.

$\pi_* \Omega_{C/S}^{\otimes n}$ is locally free sheaf on Y .

$$\text{rank} = \dim_{\kappa(q)} (\pi_* \Omega_{C/S}^{\otimes n}) \otimes \kappa(q) \stackrel{\text{by } \phi_q^0 \text{ isom.}}{=} h^0(C_q, \Omega_{C/S}^{\otimes n})$$

$$= n(2g-2) + 1 - g \quad (\text{by Riemann-Roch \& } h^1 = 0).$$

$$= (2n-1)(g-1). \quad \square$$

(b). $h^0(C_q, \Omega_{C/S}) = g$ is constant on Y , in particular locally constant.

Applying Grauert's Theorem:

25.1.5. Grauert's Theorem. — If $\pi: X \rightarrow Y$ is proper, Y is reduced and locally Noetherian, $\mathcal{F}$ is a coherent sheaf on X flat over Y , and $h^p(X_q, \mathcal{F}|_{X_q})$ is a locally constant function of $q \in Y$, then $R^p \pi_* \mathcal{F}$ is locally free, and ϕ_Z^p is an isomorphism for all $\psi: Z \rightarrow Y$.

immediately $\pi_* \Omega_{C/S}$ is locally free & ϕ_S^1 isom. $\square$

(can't get (b) using (a) because (a) needs $n > 1$
can't get (a) using (b) because Y is not reduced in (a))

If a morphism $\pi: X \rightarrow Y$ satisfies the property that the natural map $\mathcal{O}_Y \rightarrow \pi_* \mathcal{O}_X$ is an isomorphism, we say that π is **$\mathcal{O}$ -connected**. Clearly the notion of $\mathcal{O}$ -connectedness is local on the target, and preserved by composition.

25.1.1. IMPORTANT EXERCISE. Suppose $\pi: X \rightarrow Y$ is a proper flat morphism of locally Noetherian schemes, whose fibers satisfy $h^0(X_q, \mathcal{O}_{X_q}) = 1$. (Important remark: this is satisfied if π has geometrically connected and geometrically reduced fibers, by §11.4.7.) Show that π is $\mathcal{O}$ -connected. (Hint: consider ϕ_q^0)

$$(25.1.8.1) \quad \mathcal{O}_Y \otimes \kappa(q) \longrightarrow (\pi_* \mathcal{O}_X) \otimes \kappa(q) \xrightarrow{\phi_q^0} H^0(X_q, \mathcal{O}_{X_q}) \xleftarrow{\sim} \kappa(q).$$

The composition is surjective, hence ϕ_q^0 is surjective, hence it is an isomorphism by the Cohomology and Base Change Theorem 25.1.6(i). Then by the Cohomology and Base Change Theorem 25.1.6(ii), $\pi_* \mathcal{O}_X$ is locally free, thus of rank 1. Perhaps use Nakayama's Lemma to show that a map of invertible sheaves $\mathcal{O}_Y \rightarrow \pi_* \mathcal{O}_X$ that is a surjection on fibers over points (with reduced structure) is necessarily an isomorphism of sheaves. Then extract "surjection on fibers" from (25.1.8.1).

After using Cohomology and Base Change Theorem, we have $\pi_* \mathcal{O}_X$ is a line bundle on Y

and for $\mathcal{O}_Y \rightarrow \pi_* \mathcal{O}_X$, for any $q \in Y$, $\mathcal{O}_Y \otimes \kappa(q) \xrightarrow{\sim} \pi_* \mathcal{O}_X \otimes \kappa(q)$ are isomorphism.

hope to get $\mathcal{O}_{Y,q} \rightarrow (\pi_* \mathcal{O}_X)_q$ isomorphism. It is easy. $\square$

Details: if $\mathcal{O}_{Y,q} \rightarrow (\pi_* \mathcal{O}_X)_q \cong \mathcal{O}_{Y,q}$ maps 1 to a ,
the induced $\kappa(q) \xrightarrow{\sim} (\pi_* \mathcal{O}_X)_q \otimes \kappa(q)$ maps 1 to $\bar{a}$.
 $\Rightarrow a \notin \mathfrak{m}_q$. It can only be unit in $\mathcal{O}_{Y,q}$ since $\mathcal{O}_{Y,q}$ is local ring

(In Nakayama's sense: cokernel $\mathcal{O}_Y \rightarrow \pi_* \mathcal{O}_X \rightarrow \mathcal{K} \rightarrow 0$.
 $\mathcal{O}_Y \otimes \kappa(q) \rightarrow \pi_* \mathcal{O}_X \otimes \kappa(q) \rightarrow \mathcal{K} \otimes \kappa(q) \rightarrow 0 \quad \forall q$
By Nakayama $\mathcal{K}_q = 0 \quad \forall q \Rightarrow \mathcal{K} = 0$.
both sides free of rk 1 $\therefore$ surj $\Rightarrow$ isom. locally $\mathcal{O}_Y \cong \pi_* \mathcal{O}_X$)

respectively discuss when $\mathcal{M} \rightarrow \pi_* \pi^* \mathcal{M}$ & $\pi^* \pi_* \mathcal{L} \rightarrow \mathcal{L}$ to be isom.

25.1.J. EXERCISE. Suppose $\pi: X \rightarrow Y$ is a proper, $\mathcal{O}$ -connected morphism of locally Noetherian schemes. Show that if $\mathcal{M}$ is any invertible sheaf on Y , then the natural morphism $\mathcal{M} \rightarrow \pi_* \pi^* \mathcal{M}$ is an isomorphism. In particular, we can recover $\mathcal{M}$ from $\pi^* \mathcal{M}$ by applying the pushforward π_* . (Your proof will work more generally when $\mathcal{M}$ is locally free of finite rank.)

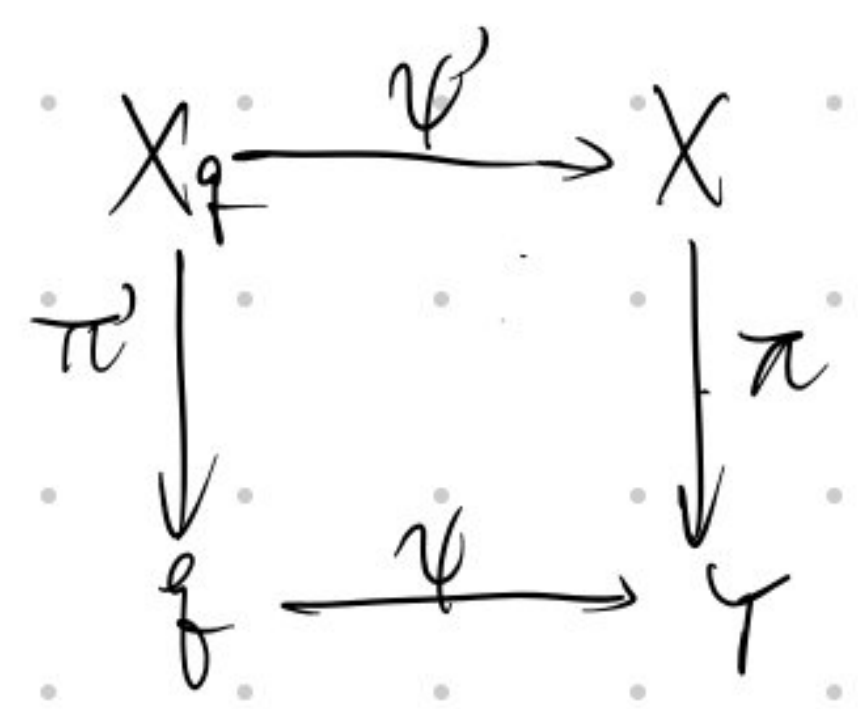
pf. projection formula: $\pi_* \pi^* \mathcal{M} \cong \pi_* (\mathcal{O}_X \otimes \mathcal{M}) \cong \mathcal{O}_Y \otimes \mathcal{M} \cong \mathcal{M}$ The proof doesn't need any condition of $X \rightarrow Y$ and X, Y except $\mathcal{O}$ -connected

not used in the proof of 25.1.11
25.1.11. Proposition. — Suppose $\pi: X \rightarrow Y$ is a flat proper morphism of locally Noetherian schemes with geometrically connected and geometrically reduced fibers (hence $\mathcal{O}$ -connected, by Exercise 25.1.I). Suppose also that Y is reduced, and $\mathcal{L}$ is an invertible sheaf on X that is trivial on the fibers of π (i.e., $\mathcal{L}|_{X_q}$ is a trivial invertible sheaf on X_q for all $q \in Y$). Then $\pi_* \mathcal{L}$ is an invertible sheaf on Y (call it $\mathcal{M}$), and the natural map $\pi^* \mathcal{M} \rightarrow \mathcal{L}$ is an isomorphism.

pf. geometrically connected & geometrically reduced fibers + $\mathcal{L}|_{X_q}$ trivial inv. sheaf $\Rightarrow h^0(X_q, \mathcal{L}|_{X_q}) = 1$ constant on Y .

Grauert's Theorem $\Rightarrow \pi_* \mathcal{L} = \mathcal{M}$. line bundle on Y .

& $\forall q \in Y$. $\mathcal{M} \otimes k(q) \xrightarrow{\cong} H^0(X_q, \mathcal{L}|_{X_q})$
 $\psi^* \pi_* \mathcal{L} \xrightarrow{\cong} \pi'_* \psi'^* \mathcal{L}$

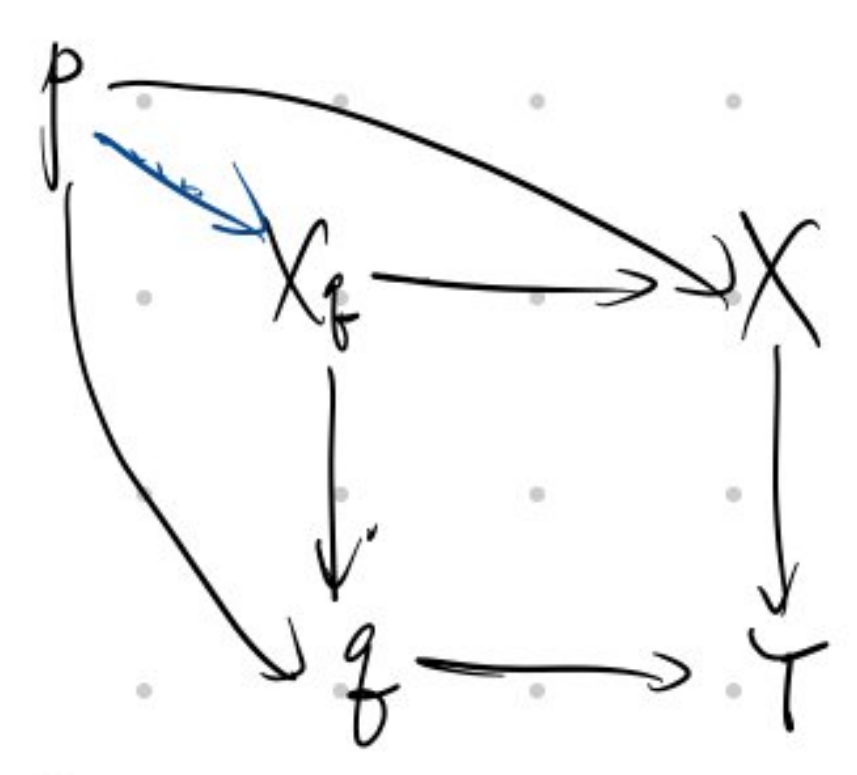


$\therefore \pi^* \psi^* \pi_* \mathcal{L} \xrightarrow{\cong} \pi^* \pi'_* \psi'^* \mathcal{L} \xrightarrow{\text{natural}} \psi'^* \mathcal{L}$
 $\pi^* \pi_* \mathcal{L}|_{X_q} \xrightarrow{\quad\quad\quad} \mathcal{L}|_{X_q}$

$\because \mathcal{L}|_{X_q} \cong \mathcal{O}_{X_q}$ on $X_q \therefore \pi'^* \pi'_* (\mathcal{L}|_{X_q}) \xrightarrow{\text{natural}} \mathcal{L}|_{X_q}$ is isomorphism

$\therefore (\pi^* \pi_* \mathcal{L})|_{X_q} \rightarrow \mathcal{L}|_{X_q}$ isom on X_q . $\forall q \in Y$

$\forall p \in X$. maps to q .



$\pi^* \pi_* \mathcal{L} \rightarrow \mathcal{L} \rightarrow \mathcal{K} \rightarrow 0$

From above. $\mathcal{K}|_{X_q} = 0$. $\therefore \mathcal{K} \otimes k(p) = 0$. $\forall p \in X$.

By Nakayama $\mathcal{K} = 0$. $\therefore \pi^* \pi_* \mathcal{L} \rightarrow \mathcal{L}$ surj.

Since $\pi_* \mathcal{L}$ line bundle on Y . it is a morphism between line bundles. hence surj. $\Leftrightarrow$ isom by easy commutative alg. □

25.1.12. Projective bundles.

can be removed. ref: 25.2.10.

25.1.K. EXERCISE. Let X be a locally Noetherian scheme, and let $\text{pr}_1: X \times \mathbb{P}^n \rightarrow X$ be the projection onto the first factor. Suppose $\mathcal{L}$ is a line bundle on $X \times \mathbb{P}^n$, whose degree on every fiber of pr_1 is zero. Use the Cohomology and Base Change Theorem 25.1.6 to show that $(\text{pr}_1)_*\mathcal{L}$ is an invertible sheaf on X . Show that the natural map $\text{pr}_1^*((\text{pr}_1)_*\mathcal{L}) \rightarrow \mathcal{L}$ is an isomorphism, as at the end of proof of Proposition 25.1.11.

pf. pr_1 is flat, proper morphism.

$\forall q \in X$. By the condition $H^1(\mathbb{P}^n_q, \mathcal{L}|_{\mathbb{P}^n_q}) \cong H^1(\mathbb{P}^n, \mathcal{O}) = 0$.

By 25.1.D. ϕ_q° isomorphism $\forall q \in X$, i.e. $\text{pr}_{1,*}\mathcal{L} \otimes k(q) \xrightarrow{\cong} H^0(\mathbb{P}^n, \mathcal{L}|_{\mathbb{P}^n_q})$.

By Cohomology and Base Change (ii) $\text{pr}_{1,*}\mathcal{L} = \mathcal{L}^\circ_{\text{pr}_1,*}\mathcal{L}$ is locally free sheaf of rank $h^0(\mathbb{P}^n, \mathcal{L}|_{\mathbb{P}^n_q}) = 1$.

Still have $\mathcal{L}|_{\mathbb{P}^n_q}$ trivial on $\mathbb{P}^n_q$. $\therefore$ use the proof in 25.1.11

(I don't repeat the proof here) $\Rightarrow \text{pr}_1^*\text{pr}_{1,*}\mathcal{L} \xrightarrow{\cong} \mathcal{L} \quad \square$

(The proof can be generalised to $\mathbb{P}^n$ -bundle $\pi: Z \rightarrow X$.

& $\mathcal{L}$ on Z such that $\mathcal{L}|_{Z_q} \cong \mathcal{O}_{\mathbb{P}^n_q}$ with $Z_q \cong \mathbb{P}^n \times q$)

25.1.L. EXERCISE. Suppose X is a connected Noetherian scheme. Show that $\text{Pic}(X \times \mathbb{P}^n) \cong \text{Pic } X \times \mathbb{Z}$. Hint: the map $\text{Pic } X \times \text{Pic } \mathbb{P}^n \rightarrow \text{Pic}(X \times \mathbb{P}^n)$ is given by $(\mathcal{L}, \mathcal{O}(m)) \mapsto \text{pr}_1^*\mathcal{L} \otimes \text{pr}_2^*\mathcal{O}(m)$, where $\text{pr}_1: X \times \mathbb{P}^n \rightarrow X$ and $\text{pr}_2: X \times \mathbb{P}^n \rightarrow \mathbb{P}^n$ are the projections from $X \times \mathbb{P}^n$ to its factors. (The notation $\boxtimes$ is often used for this construction, see §15.2.10.)

pf. Claim (naturally): $\forall \mathcal{M} \in \text{Pic}(X \times \mathbb{P}^n)$, $\mathcal{M}|_{Z \times \mathbb{P}^n} \cong \mathcal{O}(d(q))$, for some $d(q) \in \mathbb{Z}$.

Then the function $X \rightarrow \mathbb{Z}: q \mapsto d(q)$ is locally constant!

reason: $\forall q_0 \in X$, $d_0 := d(q_0)$. choose $N \gg 0$.

By upper-semicontinuous theorem, $h^0(\mathbb{P}^n_{X,q}, \mathcal{M} \otimes \text{pr}_2^*\mathcal{O}(N)|_{\mathbb{P}^n_q})$ is upper-semicontinuous, in particular $\exists U \ni q_0$ s.t. $\forall q \in U$,

$$h^0(\mathbb{P}^n_q, \mathcal{M} \otimes \text{pr}_2^*\mathcal{O}(N)|_{\mathbb{P}^n_q}) \leq h^0(\mathbb{P}^n_{q_0}, \mathcal{M} \otimes \text{pr}_2^*\mathcal{O}(N)|_{\mathbb{P}^n_{q_0}})$$

$$\therefore d(q) + N \leq d(q_0) + N$$

$\forall$ by N sufficiently large.

depending on the two h^0 are both 0 and both degree are negative & can't compare.

Do the same thing for $\mathcal{M}^\vee$, get $-d(q) \leq -d(q_0)$.

$\therefore d(q)$ is locally constant on X .

By the connectedness of X . $X \rightarrow Z: q \mapsto \deg$ is hence constant, denote by d .

$$\begin{aligned} \text{Consider } \text{Pic } X \times \text{Pic } \mathbb{P}^n &\xrightleftharpoons[\Psi]{\Phi} \text{Pic}(X \times \mathbb{P}^n) \\ (\mathcal{L}, \mathcal{O}(m)) &\xrightarrow{\Psi} \text{pr}_1^* \mathcal{L} \otimes \text{pr}_2^* (\mathcal{O}(m)) \\ (\text{pr}_1^* (\mathcal{M} \otimes \text{pr}_2^* \mathcal{O}(-d)), \mathcal{O}(-d)) &\longleftarrow \mathcal{M} \end{aligned}$$

$\Psi \circ \Phi = \text{id}$ by 25.1.J and important remark in 25.1.I.
 $\Phi \circ \Psi = \text{id}$ by 25.1.K $\square$

(Remark: can be generalised to: $Z \xrightarrow{\mathbb{P}^n\text{-bundle}} X$ with X connected noetherian.
 by $\mathbb{P}(\mathcal{E}) \rightarrow X$ for a vector bundle $\mathcal{E}$ on X

Fact: can be removed.

(reduced can be removed in my proof)

25.1.M. EXERCISE. Suppose $\pi: X \rightarrow Y$ is a projective flat morphism over a Noetherian integral scheme, all of whose geometric fibers are isomorphic to $\mathbb{P}^n$ (over the appropriate field). Show that π is a projective bundle if and only if there is an invertible sheaf $\mathcal{L}$ on X that restricts to $\mathcal{O}(1)$ on all the geometric fibers. (One direction is clear: if it is a projective bundle, then it has a $\mathcal{O}(1)$ which comes from the projectivization, see Exercise 17.2.D. In the other direction, the candidate vector bundle is $\pi_* \mathcal{L}$. Show that it is indeed a locally free sheaf of the desired rank. Show that its projectivization is indeed $\pi: X \rightarrow Y$.)

pf " $\Rightarrow$ " if $X = \mathbb{P}\mathcal{E}$. $\mathcal{O}_X(1)$ can be constructed and restricts to $\mathcal{O}(1)$ on every fibers.

" $\Leftarrow$ " if $\mathcal{L}$ on X such that by $X_{\bar{q}} \cong \mathbb{P}_{k(q)}^n$, $\mathcal{L}|_{X_{\bar{q}}} \cong \mathcal{O}_{\mathbb{P}^n}(1) \quad \forall q \in Y$.

See the link in the next page

$$\therefore h^0(X_{\bar{q}}, \mathcal{L}|_{X_{\bar{q}}}) = h^0(X_{\bar{q}}, \mathcal{O}_{\mathbb{P}^n}(1)) = n+1$$

Y integral (especially reduced). Grauert's Theorem.

$\pi_* \mathcal{L} = \mathcal{F}$ locally free sheaf on Y of rank $n+1$ & ϕ_q^0 isom., $\phi_{\bar{q}}^0$ isom.

Claim: $\pi^* \pi_* \mathcal{L} \rightarrow \mathcal{L}$ is surjective

reason: (essentially similar to the proof in 25.1.1)

$$\begin{array}{ccc} \forall q \in Y & (\pi^* \pi_* \mathcal{L})|_{X_{\bar{q}}} & \xrightarrow[\text{by } \phi_q^0]{\cong} \pi^* \pi_* (\mathcal{L}|_{X_{\bar{q}}}) \xrightarrow[\text{by } X_{\bar{q}} \cong \mathbb{P}^n]{\text{surj.}} \mathcal{L}|_{X_{\bar{q}}} \\ X_{\bar{q}} & \xrightarrow{\quad} X & \\ \pi|_{X_{\bar{q}}} \downarrow \quad \downarrow \pi & & \\ \bar{q} & \xrightarrow{\psi} Y & \end{array}$$

Similar discussion $\Rightarrow$ surjective.

universal property of proj. bundle $\Rightarrow \exists X \xrightarrow{f} \mathbb{P}_Y(\pi_* \mathcal{L})$ induced by $\pi^* (\pi_* \mathcal{L}) \rightarrow \mathcal{L}$, with $f^* \mathcal{O}_{\mathbb{P}^n}(1) \cong \mathcal{L}$.

Additions We can replace the red-framed step by:

$$H^1(X_q, \mathcal{L}|_{X_q}) = H^1(X_{\bar{q}}, \mathcal{L}|_{X_{\bar{q}}}) = 0. \quad \text{by } \mathcal{L}|_{X_{\bar{q}}} \cong \mathcal{O}_{\mathbb{P}^n}^{n+1}.$$

By cohomology and Base Change (i) $R^1\pi_{\text{red}}\mathcal{L}|_q \xrightarrow{\cong} H^1(X_q, \mathcal{L}|_{X_q}) = 0.$

By Nakayama. $R^1\pi_{\text{red}}\mathcal{L} = 0$. In particular $R^1\pi_{\text{red}}\mathcal{L} = 0$ loc. free

By Cohomology and Base Change (ii) ϕ_q^0 isom. $\forall q \in Y$.

By C & BC⁽ⁱⁱⁱ⁾ again. π_{red} is loc. free of rank $h^0 = n+1$

The proof above implies that we don't need Y to be reduced!

now want to show f is isomorphism.

Remark: Generally $X \xrightarrow{\pi} Y$ with $\mathcal{E}$: vector bundle over Y
 $\mathcal{L}$: line bundle over X . $\pi^*\mathcal{E} \rightarrow \mathcal{L}$
 in the construction of $X \xrightarrow{f} \mathbb{P}_Y(\mathcal{E})$.
 $\forall K \xrightarrow{q} Y$, $f_q: X_q \rightarrow (\mathbb{P}_Y(\mathcal{E}))_q = \mathbb{P}_{K(q)}^{rk\mathcal{E}-1}$
 is actually " $x \mapsto [s_0(x), \dots, s_{rk\mathcal{E}}(x)]$ "
 you can see this clearly from assuming $Y = k$.

$$\mathcal{O}_{X_q}^{rk\mathcal{E}} \cong (\pi^*\mathcal{E})|_{X_q} \rightarrow \mathcal{L}|_{X_q}$$

$$(e_0, \dots, e_{rk\mathcal{E}}) \mapsto (s_0, \dots, s_{rk\mathcal{E}})$$

(under our situation! i.e. $\pi^*\mathcal{L} = \mathcal{E}$.
 $\mathcal{L}|_{X_q} \cong \mathcal{O}(1)$.)

By the remark $f_q: X_q \rightarrow (\mathbb{P}_Y(\mathcal{E}))_q$ are isomorphisms. $q \in Y$.

$\therefore f_q: X_q \rightarrow (\mathbb{P}_Y(\mathcal{E}))_q$ isomorphism by fibre descent.

(In essence $\bar{q} \rightarrow q$ faithfully flat)
 more essentially $A \otimes_k L = 0 \Leftrightarrow A = 0$
 with L/k field extn.

$\therefore X \xrightarrow{f} \mathbb{P}_Y(\pi^*\mathcal{L})$ is set theoretically bijection

$\forall x \in X, \hookrightarrow z \in \mathbb{P}_Y(\pi^*\mathcal{L})$. Goal: $\mathcal{O}_{Z,z} \xrightarrow{\cong} \mathcal{O}_{X,x}$.

f is proper. by fiber isom. it is quasifm. $\therefore f$ finite.

$\therefore X_q \xrightarrow{\cong} \mathbb{P}_Y(\pi^*\mathcal{L})_q$ are isom. for $\begin{matrix} X \\ \downarrow \\ X_q \end{matrix} \begin{matrix} \nearrow \\ \downarrow \end{matrix} \begin{matrix} \mathbb{P}_Y(\pi^*\mathcal{L}) \\ \downarrow \end{matrix} \begin{matrix} \mathbb{P}_Y(\pi^*\mathcal{L})_q \end{matrix}$ $q \in Y$

$$\mathcal{O}_{Z,z} / m_q(\mathcal{O}_{Z,z}) \xrightarrow{\cong} \mathcal{O}_{X,x} / m_q(\mathcal{O}_{X,x})$$

By $- \otimes_{k(q)}$ right exact, + Nakayama. $\text{Coker}(\mathcal{O}_{Z,z} \rightarrow \mathcal{O}_{X,x}) = 0$.

i.e. $\mathcal{O}_{Z,z} \rightarrow \mathcal{O}_{X,x}$ surj.

write $0 \rightarrow I \rightarrow \mathcal{O}_{Z,z} \rightarrow \mathcal{O}_{X,x} \rightarrow 0$.

$\therefore X \xrightarrow{\pi} Y$ flat. $\therefore \text{Tor}_1^{\mathcal{O}_{Y,z}}(k(q), \mathcal{O}_{X,x}) = 0$.

$$\therefore 0 \rightarrow I \otimes_{k(q)} \rightarrow (\mathcal{O}_{Z,z} \otimes_{k(q)}) \xrightarrow{\cong} (\mathcal{O}_{X,x} \otimes_{k(q)}) \rightarrow 0$$

$\therefore$ By Nakayama. $I = 0$.

$$\therefore \mathcal{O}_{Z,z} \xrightarrow{\cong} \mathcal{O}_{X,x} \quad \square$$

25.3

25.3.3 Grassmannian "for the third time"

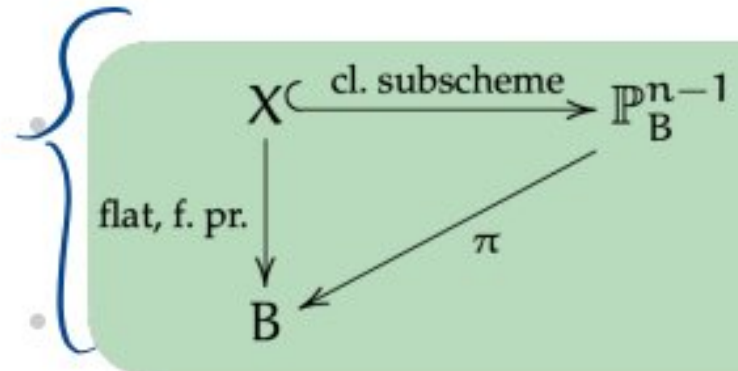
Recall: (cf. 25.1.4).

$G(k, n): \text{Sch}^{\text{pt}} \rightarrow \text{Sets}$ $\text{rk} = k$ locally free

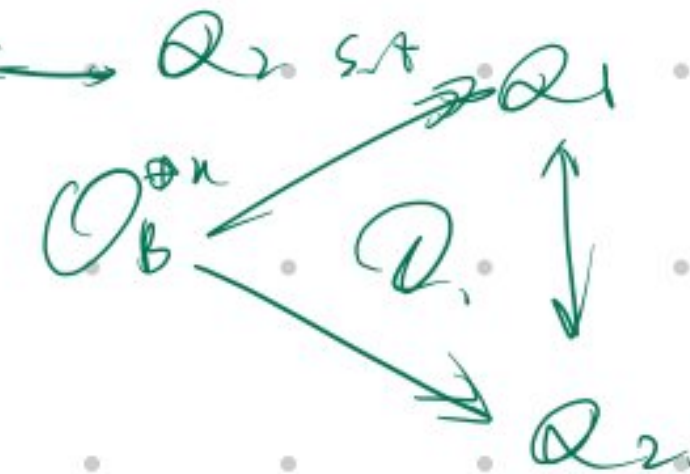
$B \mapsto \{ \mathcal{O}_B^{\oplus n} \rightarrow \mathcal{Q} \} / \sim$ see $\mathcal{O}_B^{\oplus n} \rightarrow \mathcal{Q}_1$ and $\mathcal{O}_B^{\oplus n} \rightarrow \mathcal{Q}_2$ as the same if $\exists$ isom $\mathcal{Q}_1 \xrightarrow{\sim} \mathcal{Q}_2$ s.t.

Def $G'(k, n): \text{Sch}^{\text{pt}} \rightarrow (\text{Sets})$

$B \mapsto$



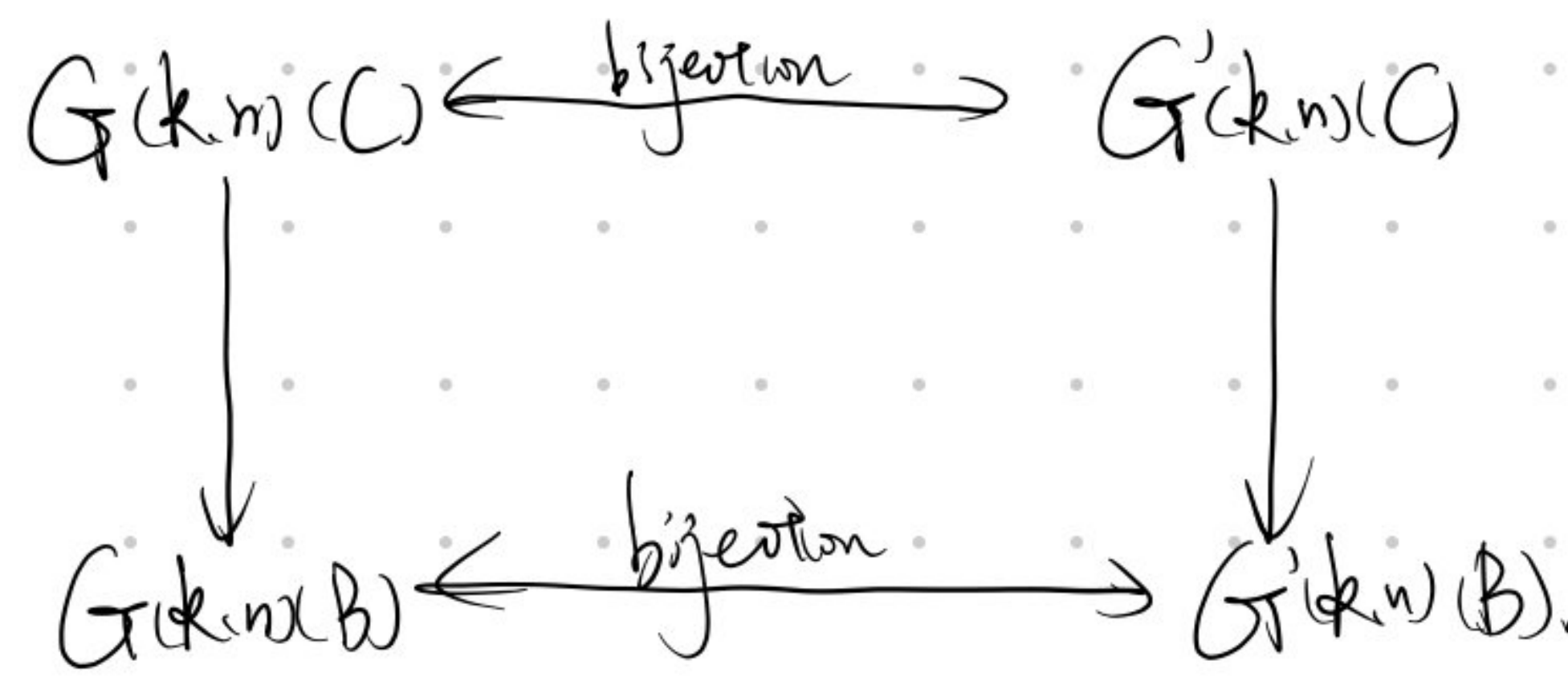
$\forall b \in B$, we have $X_b \hookrightarrow \mathbb{P}^{n-1}_{k(b)}$ $\mathbb{P}^{k-1}_{k(b)} \hookrightarrow \mathbb{P}^{n-1}_{k(b)}$ linearly embedded!



25.3.4. Theorem. — The functor $G'(k, n)$ is represented by $G(k, n)$.

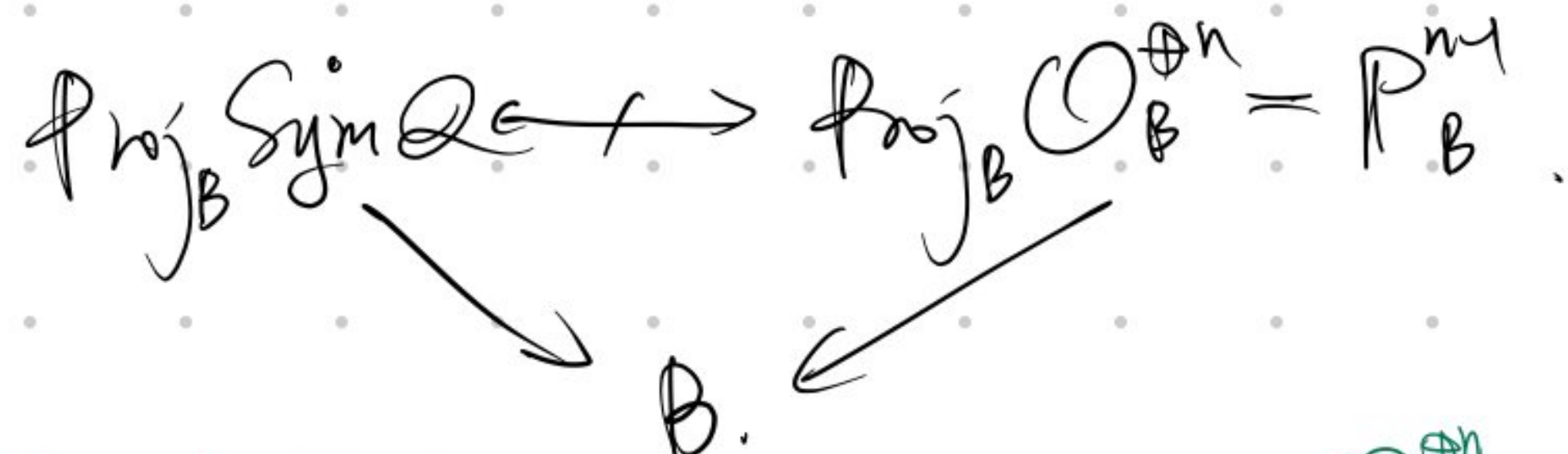
pf. i.e. showing $\forall B \in \text{Sch}$, we can construct natural bijections, i.e.

$\forall B \rightarrow C$ morphism of schemes, we have:



$G(k, n)(B) \rightarrow G'(k, n)(B)$.

If we have $\mathcal{O}_B^{\oplus n} \rightarrow \mathcal{Q}$, then we have



Remark: if affine locally $\text{Spec } A = V \subseteq B$, with $\mathcal{O}_V^{\oplus n} \rightarrow \mathcal{Q}|_V \cong \mathcal{O}_V^{\oplus k}$ $e_i \mapsto \sum a_{ij} \hat{e}_i$

Then locally the above morphism is:



By the surjectivity, $\bigcup D_+(\sum a_{ij} y_j) = P^{k-1}_A = \text{Proj } k[y_1, \dots, y_k]$.

$D_+(\sum a_{ij} y_j) \hookrightarrow D_+(x_i)$ by $k[\frac{x_1}{x_i}, \dots, \frac{x_n}{x_i}] \hookrightarrow k[\frac{y_1}{\sum a_{ij} y_j}, \dots, \frac{y_k}{\sum a_{ij} y_j}]$ $\frac{x_j}{x_i} \mapsto \sum_m a_{jm} \frac{y_m}{\sum a_{ij} y_j}$

If $A = k$, then $[a_1, \dots, a_k] \mapsto [\sum a_{ij} y_j, \dots, \sum a_{ij} y_j]$.

The construction is functorial because base change.

$G(k, m)(B) \longrightarrow G(k, n)(B)$. (we will use cohomology and base change here)

If we have $X \xrightarrow{i} \mathbb{P}_B^m$ in $G(k, n)(B)$.

$$0 \longrightarrow \mathcal{I}_X \longrightarrow \mathcal{O}_{\mathbb{P}_B^m} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

$$0 \longrightarrow \mathcal{I}_X(1) \longrightarrow \mathcal{O}_{\mathbb{P}_B^m}(1) \longrightarrow \mathcal{O}_{\mathbb{P}_B^m}(1) \otimes_{\mathcal{O}_{\mathbb{P}_B^m}} \mathcal{O}_X \longrightarrow 0.$$

Notice that $\forall b \in B$, by $X_b \hookrightarrow \mathbb{P}_{k(b)}^m$ always linear embedding.

$$\begin{aligned} \text{Detail: } X_b &\xrightarrow{i_b} \mathbb{P}_{k(b)}^m \xrightarrow{\gamma} b \quad (\mathcal{O}_{\mathbb{P}_{k(b)}^m}^{\otimes m} / \mathcal{I}_{X_b}^{\otimes m})|_{\mathbb{P}_{k(b)}^m} = \gamma^*(i_* \mathcal{O}_X \otimes \mathcal{O}_{\mathbb{P}_B^m}^{\otimes m}(1)) \\ &\downarrow \quad \downarrow \quad \downarrow \\ X &\xrightarrow{i} \mathbb{P}_B^m \xrightarrow{\pi} B \end{aligned}$$

$$\begin{aligned} &\cong i_{b*} \mathcal{O}_{X_b} \otimes \mathcal{O}_{\mathbb{P}_{k(b)}^m}^{\otimes m}(1) \\ &\cong i_{b*} (\mathcal{O}_{X_b} \otimes i_b^* \mathcal{O}_{\mathbb{P}_B^m}^{\otimes m}(1)) \\ &\cong i_{b*} (\mathcal{O}_{\mathbb{P}_{k(b)}^m}^{\otimes m}(1)). \end{aligned}$$

use here.

$$H^j(\mathbb{P}_{k(b)}^m, (i_* \mathcal{O}_X \otimes \mathcal{O}_{\mathbb{P}_B^m}^{\otimes m}(1))|_{\mathbb{P}_{k(b)}^m}) = H^j(\mathbb{P}_{k(b)}^m, i_{b*} \mathcal{O}_{\mathbb{P}_{k(b)}^m}^{\otimes m}(1))$$

$$\cong H^j(X_b, \mathcal{O}_{X_b}(1)) \quad \text{by } i_b \text{ closed immersion} \Rightarrow \text{affine}$$

$j=1$, we get 0, so by Cohomology and Base Change (i), & R.S.I.C.

$$R^1 \pi_* (i_* \mathcal{O}_X \otimes \mathcal{O}_{\mathbb{P}_B^m}^{\otimes m}(1)) = 0, \quad \text{in particular loc. free.}$$

by Cohomology and Base Change (ii). ϕ_b^0 is isom. for $(i_* \mathcal{O}_X \otimes \mathcal{O}_{\mathbb{P}_B^m}^{\otimes m}(1))$

by Cohomology and Base Change (ii). again. $\pi_*(i_* \mathcal{O}_X \otimes \mathcal{O}_{\mathbb{P}_B^m}^{\otimes m}(1))$ loc. free.

$$\text{Also, } 0 \longrightarrow \pi_* \mathcal{I}_X \longrightarrow \pi_* \mathcal{O}_{\mathbb{P}_B^m}(1) \longrightarrow \pi_* (\mathcal{O}_{\mathbb{P}_B^m}(1) \otimes i_* \mathcal{O}_X) \longrightarrow R^1 \pi_* \mathcal{I}_X \longrightarrow \dots$$

Claim: $\pi_* (\mathcal{O}_{\mathbb{P}_B^m}(1) \otimes i_* \mathcal{O}_X) \longrightarrow \pi_* (\mathcal{O}_{\mathbb{P}_B^m}(1) \otimes i_* \mathcal{O}_X)$ is surjective.

reason:

$$\pi_* (\mathcal{O}_{\mathbb{P}_B^m}(1) \otimes \kappa(b)) \longrightarrow \pi_* (\mathcal{O}_{\mathbb{P}_B^m}^{\otimes m} \otimes i_* \mathcal{O}_X \otimes \kappa(b))$$

$$\cong \downarrow \phi_b^0$$

$$\cong \downarrow \phi_b^0$$

$$H^0(\mathbb{P}_{k(b)}^m, \mathcal{O}_{\mathbb{P}_{k(b)}^m}^{\otimes m}(1)) \xrightarrow[\text{by linear embedding}]{\text{surj.}} H^0(\mathbb{P}_{k(b)}^m, i_{b*} \mathcal{O}_{X_b}(1))$$

$$\hookrightarrow \text{coker}(\pi_* \mathcal{O}_{\mathbb{P}_B^m}(1) \longrightarrow \pi_* (\mathcal{O}_{\mathbb{P}_B^m}(1) \otimes i_* \mathcal{O}_X)) \otimes \kappa(b) = 0.$$

By Nakayama $\text{coker} = 0$. $\therefore$ The claim holds!

$$\cdot, 0 \longrightarrow \pi_* \mathcal{I}_X \longrightarrow \pi_* \mathcal{O}_{\mathbb{P}^n_B}(1) \longrightarrow \pi_* (\mathcal{O}_{\mathbb{P}^n_B}(1) \otimes i_* \mathcal{O}_X) \longrightarrow 0. \quad (2S.3.F).$$

$$\parallel \downarrow$$

$$\mathcal{O}_B^{\oplus n}$$

Consider $G'(k,n)(B) \longrightarrow G(k,n)(B)$.

$$\left(\begin{array}{ccc} X & \xrightarrow{i} & \mathbb{P}^n_B \\ & \searrow f & \swarrow \pi \\ & B & \end{array} \right) \mapsto \left(\mathcal{O}_B^{\oplus n} \longrightarrow \pi_* (\mathcal{O}_{\mathbb{P}^n_B}(1) \otimes i_* \mathcal{O}_X) \right)$$

has been proved loc. free of rank k .

$$\cdot (G(k,n) \longrightarrow G'(k,n) \longrightarrow G(k,n)) = \text{id}_{G(k,n)}.$$

ie. To show: $\forall \mathcal{O}_B^{\oplus n} \twoheadrightarrow \mathcal{Q}$ with $\mathcal{Q}$ loc. free of $\text{rd} = k$.

$$\begin{array}{ccc} X = \text{Proj Sym } \mathcal{Q} & \xrightarrow{i} & \mathbb{P}^n_B \\ & \searrow f & \swarrow \pi \\ & B & \end{array} \quad \text{we have}$$

$$\begin{array}{ccc} & & \mathcal{Q} \\ & \nearrow & \uparrow \cong \\ \mathcal{O}_B^{\oplus n} & & \\ & \searrow & \downarrow \\ & & \pi_* (\mathcal{O}_{\mathbb{P}^n_B}(1) \otimes i_* \mathcal{O}_X). \end{array}$$

recall: $\pi_* (\mathcal{O}_{\mathbb{P}^n_B}(1) \otimes i_* \mathcal{O}_X) \cong \pi_* i_* i^* \mathcal{O}_{\mathbb{P}^n_B}(1)$ by proj. formula

$$= f_* \mathcal{O}_X(1).$$

$$\cong \mathcal{Q}.$$

$$\cdot (G'(k,n) \longrightarrow G(k,n) \longrightarrow G'(k,n)) = \text{id}_{G(k,n)}$$

ie. To show: $\forall X \xrightarrow{i} \mathbb{P}^n_B$ we have

$$\begin{array}{ccc} & X & \xrightarrow{i} \mathbb{P}^n_B \\ & \searrow f & \swarrow \pi \\ & B & \end{array}$$

$$\begin{array}{ccc} \text{Proj}_B \text{ Sym } (\pi_* (\mathcal{O}_{\mathbb{P}^n_B}(1) \otimes i_* \mathcal{O}_X)) & \xrightarrow{\cong} & X \\ \downarrow f_* i^* \mathcal{O}_{\mathbb{P}^n_B}(1) & & \downarrow f \\ & B & \end{array}$$

Recall: the construction in 2S.1.M, $X \cong \text{Proj}_B (f_* i^* \mathcal{O}_{\mathbb{P}^n_B}(1))$. $\square$

Now: $G(k,n)$ and $G'(k,n)$ are the same thing!

25.3.5 Hypersurfaces

Def $H_{d,n} : \text{Sch}^{\text{op}} \longrightarrow \text{Sets}$

$$B \longmapsto \left\{ \begin{array}{c} X \hookrightarrow \mathbb{P}_B^{n+1} \\ \text{flat} \\ \text{finitely presented} \\ \downarrow \pi \\ B \end{array} \mid \forall b \in B, X_b \hookrightarrow \mathbb{P}_{k(b)}^{n+1} \text{ is actually a hypersurface of deg } d \right\}$$

25.3.6. Proposition. — The functor $H_{d,n}$ is represented by $\mathbb{P}^{\binom{n+d}{d}-1}$.

As with the case of the Grassmannian, one direction is easy, and the other requires cohomology and base change.

pf.

25.3.H. EASY EXERCISE. Over $\mathbb{P}^{\binom{n+d}{d}-1}$, describe a closed subscheme $\mathcal{X} \hookrightarrow \mathbb{P}^n \times \mathbb{P}^{\binom{n+d}{d}-1}$ that will be the universal hypersurface (the universal family over the moduli space). Show that $\mathcal{X}$ is flat and finitely presented over $\mathbb{P}^{\binom{n+d}{d}-1}$. (For flatness, you can use the local criterion of flatness on the source, Theorem 24.6.9, but it is possible to deal with it easily by working by hand.)

have easy to construct $\mathcal{X}$ from this idea. kpe

$$\begin{array}{ccc} \mathcal{X} \hookrightarrow \mathbb{P}^n \times \mathbb{P}^{\binom{n+d}{d}-1} & \xrightarrow{\text{pr.}} & \mathbb{P}^{\binom{n+d}{d}-1} \\ & \searrow \pi & \\ & & \mathbb{P}^{\binom{n+d}{d}-1} \end{array}$$

(Natural but not rigorous. The fibre of $\mathcal{X} \rightarrow \mathbb{P}^{\binom{n+d}{d}-1}$ at $[a_{i_1, \dots, i_d}]_{0 \leq i_1, \dots, i_d \leq n}$ is just $V_+(\sum_{(i_1, \dots, i_d)} a_{i_1, \dots, i_d} x^{i_1} \dots x^{i_d}) \subseteq \mathbb{P}_k^n$.)

- $\mathcal{X} \rightarrow \mathbb{P}^{\binom{n+d}{d}-1}$ fin. presented by pr and closed immersion are fin. presented.
- $\forall x \in \mathcal{X}, \pi(x) = s.$

Apply Thm 24.6.9 to $(\mathcal{O}_{\mathbb{P}^{\binom{n+d}{d}-1}, s}, M_s) \rightarrow (\mathcal{O}_{\mathbb{P}^n \times \mathbb{P}^{\binom{n+d}{d}-1}, x}, M_x).$

24.6.9. Theorem (slicing criterion for flatness on the source, see Figure 24.4). — Suppose $(B, n) \rightarrow (A, m)$ is a local morphism of Noetherian local rings, M is a finitely generated A -module, and $t \in m \subset A$ is a non-zerodivisor on M/nM . If M is flat over B then M/tM is flat over B .

Recall: affine locally $(B_q, qB_q) \rightarrow (A_p, pA_p)$ $\text{Spec } A \rightarrow \text{Spec } B$

$$\begin{array}{ccc} \mathcal{O}_{\text{Spec } B, [q]} & \xrightarrow{\text{pr.}} & \mathcal{O}_{\text{Spec } A, [p]} \\ \mathcal{O}_{\text{Spec } B_q/qB_q, [q]} & \xrightarrow{\text{pr.}} & \mathcal{O}_{\text{Spec } A_p/pA_p, [p]} \end{array}$$

$$\mathcal{O}_{\text{Spec } B_q/qB_q, [q]} \otimes_{\mathcal{O}_{\text{Spec } B_q/qB_q, [q]}} \mathcal{O}_{\text{Spec } A_p/pA_p, [p]} = (\mathcal{O}_{\text{Spec } B_q/qB_q, [q]} \otimes_{\mathcal{O}_{\text{Spec } B_q/qB_q, [q]}} \mathcal{O}_{\text{Spec } A_p/pA_p, [p]}) \otimes_{\mathcal{O}_{\text{Spec } B_q/qB_q, [q]}} \mathcal{O}_{\text{Spec } A_p/pA_p, [p]} \cong \mathcal{O}_{\text{Spec } A_p/pA_p, [p]} \otimes_{\mathcal{O}_{\text{Spec } A_p/pA_p, [p]}} \mathcal{O}_{\text{Spec } A_p/pA_p, [p]} \cong \mathcal{O}_{\text{Spec } A_p/pA_p, [p]}$$

$\frac{M}{nM}$ corresponds to $\mathcal{O}_{(\mathbb{P}^n \times \mathbb{P}^{\binom{n+d}{d}-1})_s, x} = \mathcal{O}_{\frac{\mathbb{P}^n}{\text{integral}}, x}$ is integral

t corresponds to $\mathcal{X}_s = V_+(t) \subseteq \mathbb{P}_{k(s)}^n$

$$t = \sum_{i_1, \dots, i_d \in K(s)} a_{i_1, \dots, i_d}(s) \cdot x^{i_1} \dots x^{i_d}$$

must exist one nonzero $(\because \mathcal{O}_{\mathbb{P}^{n+1}(b)} \text{ g.g.})$
 $\therefore \mathcal{X} \rightarrow \mathbb{P}^{n+1}$ is faithfully flat by 24.1.6. $N = \binom{nd}{d}$

• $\mathrm{Hom}_{\mathrm{Sch}}(B, \mathbb{P}^{n+1}) \rightarrow \mathrm{Hom}_{\mathrm{Sch}}(B, B)$

$$(B \rightarrow \mathbb{P}^{n+1}) \mapsto \left(\begin{array}{ccc} \mathcal{X}_{\mathbb{P}^{n+1}} B & \hookrightarrow & \mathbb{P}^n \times B \\ & \searrow & \downarrow \\ & & B \end{array} \right)$$

• $\mathrm{Hom}_{\mathrm{Sch}}(B, \mathbb{P}^{n+1}) \rightarrow \mathrm{Hom}_{\mathrm{Sch}}(B, \mathbb{P}^{n+1})$

From $X \hookrightarrow \mathbb{P}^n \times B$
 $\downarrow \pi$
 B

$$\begin{aligned} 0 \rightarrow \mathcal{I}_X \rightarrow \mathcal{O}_{\mathbb{P}^n_B} \rightarrow \mathcal{O}_X \rightarrow 0 \\ \rightarrow \mathcal{I}_X(d) \rightarrow \mathcal{O}_{\mathbb{P}^n_B}(d) \rightarrow \mathcal{O}_X(d) \rightarrow 0 \end{aligned}$$

As in the proof before $\forall b \in B$

$$H^1(\mathbb{P}^n_{k(b)}, \mathcal{O}_X(d)|_{\mathbb{P}^n_{k(b)}}) = H^1(X_b, \mathcal{O}_{X_b}(d)) = 0$$

(This is because $0 \rightarrow \mathcal{I}_{X_b} \cong \mathcal{O}_{\mathbb{P}^n_{k(b)}}(d-d) \rightarrow \mathcal{O}_{\mathbb{P}^n_{k(b)}}(d) \rightarrow \mathcal{O}_{X_b}(d) \rightarrow 0$)
 and apply long exact sequence

By 25.1.C $R^1 \pi_* (\mathcal{O}_X(d)) = 0$

By Cohomology and Base Change (13) ϕ^0 surj, again. $\pi_* \mathcal{O}_X(d)$ loc. free of rank $h^0(\mathcal{O}_{X_b}(d))$

$$= h^0(\mathcal{O}_{\mathbb{P}^n}(d)) - h^0(\mathcal{O}_{\mathbb{P}^n}) = \binom{nd}{d} - 1$$

Also as in the proof before

$$0 \rightarrow \pi_* \mathcal{I}_X(d) \rightarrow \pi_* \mathcal{O}_{\mathbb{P}^n_B}(d) \rightarrow \pi_* \mathcal{O}_X(d) \rightarrow 0$$

The surjectivity is by: $\forall b \in B$

$$\begin{array}{ccc} (\pi_* \mathcal{O}_{\mathbb{P}^n_B}(d)) \otimes_{k(b)} & \xrightarrow{\quad} & (\pi_* \mathcal{O}_{X_b}(d)) \otimes_{k(b)} \\ \cong \downarrow & & \downarrow \cong \\ H^0(\mathcal{O}_{\mathbb{P}^n_{k(b)}}(d)) & \xrightarrow{\text{surj.}} & H^0(\mathcal{O}_{X_b}(d)) \end{array}$$

and analyse coker. using Nakayama as before.

$\pi_* \mathcal{I}_X(d)$ is loc. free of $rk = \binom{n+d}{d} - \left(\binom{n+d}{d} - 1 \right) = 1$.

consider $\mathcal{O}_B^{\oplus \binom{n+d}{d}} \longrightarrow (\pi_* \mathcal{I}_X(d))^\vee$

15.2.2. Important Theorem. — For a fixed scheme X , morphisms $X \rightarrow \mathbb{P}^n$ are in bijection with the data $(\mathcal{L}, s_0, \dots, s_n)$, where $\mathcal{L}$ is an invertible sheaf and $s_0, \dots, s_n$ are sections of $\mathcal{L}$ with no common zeros (i.e., $(s_0, \dots, s_n)$ is base-point-free), up to isomorphism of these data.

$$\leadsto B \longrightarrow \mathbb{P}^{\binom{n+d}{d}-1}$$

$$\bullet \text{ Hom}_{\text{Sch}}(B, \mathbb{P}^n) \xrightarrow{\quad} \text{Hden}(B) \xrightarrow{\quad} \text{Hom}_{\text{Sch}}(B, \mathbb{P}^n)$$

id

$$B \xrightarrow{f} \mathbb{P}^n \text{ with } f^* \mathcal{O}_{\mathbb{P}^n}(1) = \mathcal{L}$$

$$\begin{array}{ccccc} \mathbb{A}^n_{\mathbb{P}^n} B \hookrightarrow \mathbb{P}^n_B & \xrightarrow{\pi} & B & & \\ \downarrow \square & \downarrow g & \downarrow \square & \downarrow f & \\ \mathbb{A}^n \hookrightarrow \mathbb{P}^n \times \mathbb{P}^n & \xrightarrow{\eta} & \mathbb{P}^n & & \\ & \text{flat} & & & \end{array}$$

24.2.2. Pulling back quasicoherent ideal sheaves of flat closed subschemes by arbitrary morphisms (promised in §24.1.10). Suppose

using 24.2.11

$$\begin{array}{ccc} W & \xrightarrow{\alpha} & X \\ \downarrow & & \downarrow \\ Y & \xrightarrow{\beta} & Z \end{array}$$

is a fibered product, and $V \hookrightarrow X$ is a closed subscheme. Then $Y \times_Z V$ is a closed subscheme of W (§10.2.2). There are two possible senses in which $\mathcal{I}_{V/X}$ can be “pulled back” to W : as a quasicoherent sheaf $\alpha^* \mathcal{I}_{V/X}$, and as the ideal of the “pulled back” closed subscheme $\mathcal{I}_{Y \times_Z V/W}$. As pointed out in Remark 14.5.9, these are not necessarily the same, but they are the same if β is flat. We now give another important case in which they are the same.

24.2.H. EXERCISE. If V is flat over Z (with no hypotheses on β), show that $\alpha^* \mathcal{I}_{V/X} \cong \mathcal{I}_{Y \times_Z V/W}$. Hint: Exercise 24.2.D.

$$\mathcal{I}_{\mathbb{A}^n_B / \mathbb{P}^n_B} \cong g^* \mathcal{I}_{\mathbb{A}^n / \mathbb{P}^n \times \mathbb{P}^n} = g^* \mathcal{O}_{\mathbb{P}^n \times \mathbb{P}^n}(-d, -1) = \mathcal{O}_{\mathbb{P}^n}(-d) \otimes \pi^* \mathcal{L}^\vee$$

$$\pi_* \mathcal{I}_{\mathbb{A}^n_B / \mathbb{P}^n_B}(d) \cong \pi_* \pi^* \mathcal{L}^\vee \cong \mathcal{L}^\vee \text{ by 25.1.1}$$

$$\text{the same: } \eta_* \mathcal{I}_{\mathbb{A}^n / \mathbb{P}^n \times \mathbb{P}^n}(d) \cong \eta_* \eta^* \mathcal{O}_{\mathbb{P}^n \times \mathbb{P}^n}(-d, -1) \cong \mathcal{O}_{\mathbb{P}^n}(-d)$$

$$0 \longrightarrow \mathcal{I}_{\mathbb{A}^n / \mathbb{P}^n \times \mathbb{P}^n}(d) \longrightarrow \mathcal{O}_{\mathbb{P}^n \times \mathbb{P}^n}(d, 0) \longrightarrow \dots \longrightarrow 0$$

$$\mathcal{O}_{\mathbb{P}^n \times \mathbb{P}^n}(0, -1)$$

$$\text{locally } f \longmapsto f \otimes F$$

$\deg(0, -1) \quad \deg(0, 0) \quad \deg(d, 1)$

$$\leadsto 0 \longrightarrow \mathcal{O}_{\mathbb{P}^n}(-d) \longrightarrow \mathcal{O}_{\mathbb{P}^n}^{\oplus N} \longrightarrow \dots \longrightarrow 0$$

locally $f \mapsto \{f \cdot a_{i1}, \dots, a_{iN}\}$

$\leadsto \mathcal{O}_{\mathbb{P}^{n-1}}^{\oplus n} \rightarrow \mathcal{O}_{\mathbb{P}^{n-1}}(1)$ just corresponds to $\mathbb{P}^{n-1} \xrightarrow{\text{id}} \mathbb{P}^{n-1}$.
 $\leadsto \mathcal{O}_B^{\oplus n} \rightarrow \mathcal{L}$ by f^* just corresponds to $B \xrightarrow{f} \mathbb{P}^{n-1}$.

$\bullet \text{ Hom}(B) \rightarrow \text{Hom}_{\text{Sch}}(B, \mathbb{P}^{n-1}) \rightarrow \text{Hom}(B)$
 $\quad \quad \quad \text{id}$

For $X \hookrightarrow \mathbb{P}_B^n$
 $\quad \searrow \quad \swarrow \pi$
 $\quad \quad B$

where $B \xrightarrow{f} \mathbb{P}^{n-1}$ is constructed by $\mathcal{O}_B^{\oplus n} \rightarrow (\pi_* \mathcal{I}_{X/\mathbb{P}_B^n}(d))^\vee$.

Hope to prove $X \cong X \times_{\mathbb{P}^{n-1}} B$.

$\mathcal{I}_{X \times_{\mathbb{P}^{n-1}} B / \mathbb{P}_B^n} \cong \mathcal{O}_{\mathbb{P}_B^n}(d) \otimes \pi^* (\pi_* \mathcal{I}_{X/\mathbb{P}_B^n}(d))^\vee$

By 24.2.11 again

$(\mathcal{I}_{X/\mathbb{P}_B^n}(d))|_{\mathbb{P}_B^n} \cong \mathcal{I}_{X_B/\mathbb{P}_B^n}(d)$
 $\cong \mathcal{O}_{\mathbb{C}}(d)$
 $\cong \mathcal{O}_{\mathbb{P}_B^n}^n$
 trivial

natural $\cong$

By 25.1.11

$\cong \mathcal{O}_{\mathbb{P}_B^n}(d) \otimes \mathcal{I}_{X/\mathbb{P}_B^n}(d) \cong \mathcal{I}_{X/\mathbb{P}_B^n}$ $\square$

naturally

naturally

$\mathcal{O}_{\mathbb{P}_B^n}$